

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

First Moment Method

• A, B events

$$P(A \cup B) \leq P(A) + P(B).$$

• More generally

$$P(A_1 \cup \dots \cup A_n) \leq \sum_{i=1}^n P(A_i)$$

"union bound".

Ramsey

Defⁿ

$R(k, l) = \min \{ n : \text{every red/blue col. of edges of } K_n \text{ contains red } K_k \text{ or blue } K_l \}$.



$\Rightarrow R(3, 3) > 5$. In fact $R(3, 3) = 6$. (Cere bcrly).

$$R(4, 4) = 18$$

$$\underset{\uparrow 1989}{43} \leq R(5, 5) \leq \underset{\uparrow}{46}$$

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$$(1+o(1)) \frac{\sqrt{25}}{e} \cdot 2^{5/2}$$

$$\leq R(s, s) \underset{\substack{\uparrow \\ 2023}}{\leq} (4-\epsilon)^s \quad \epsilon = 2^{-7}$$

Thm

Let $n, k \in \mathbb{N}$ s.t. $\binom{n}{k} 2^{1 - \binom{k}{2}} < 1$

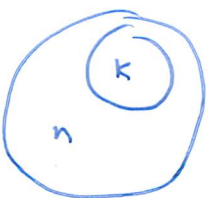
then $R(k, k) > n$.

Pf:

Take K_n , colour edges R/B, each R w. prob. $1/2$, B w. prob. $1/2$, colours of all edges indep.

Note $\binom{n}{k}$ copies of K_k in K_n

Let A_i event i^{th} copy monochromatic for $i = 1, \dots, \binom{n}{k}$.



$$P(\exists \text{ monoch. } K_k)$$

$$= P(A_1 \cup \dots \cup A_{\binom{n}{k}})$$

$$\leq \sum_{i=1}^{\binom{n}{k}} P(A_i) = \binom{n}{k} 2^{1 - (\frac{k}{2})} < 1$$

$\uparrow$
by assumption.

$$\Rightarrow P(\text{no monoch. } K_k) > 0.$$

$$\Rightarrow \exists \text{ a } R/B \text{ colouring of } K_n, \text{ s.t. no monoch. } K_k$$

$$\Rightarrow R(k, k) > n.$$

□

Cor $R(k, k) \geq 2^{k/2} \quad \forall k \geq 3.$

Pf Note $\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{n(n-1)\dots 1}{k(k-1)\dots 1} \leq \frac{n^k}{k!}$

let $n = \lfloor 2^{k/2} \rfloor$, $\binom{n}{k} 2^{1 - (\frac{k}{2})} \leq \frac{(2^{k/2})^k}{k!} \cdot 2^1 \cdot 2^{-k(k-1)/2}$

but $2^{k/2} + 1 < k!$, thus result follows from prev. Th

□

Defⁿ X r.v. it's 'first moment' is $E(X)$.

Recall X, Y r.v., $\lambda \in \mathbb{R}$ (constant)

$$\Rightarrow \bullet E[X+Y] = E[X] + E[Y]$$

$$\bullet E[\lambda X] = \lambda \cdot E[X]$$

Defⁿ A event then 'it's indicator function' I_A is r.v.
$$I_A = \begin{cases} 1 & \text{when } A \text{ holds} \\ 0 & \text{o.w.} \end{cases}$$

Note $A_1, \dots, A_n$ events, $I_i = I_{A_i}$ then r.v. X

$$\bullet X = \sum_{i=1}^n I_i = \# \text{ events that occur.}$$

$$\bullet E[X] = \sum_{i=1}^n E[I_i] = \sum_{i=1}^n P[A_i].$$

$$\bullet P(X \geq E[X]) > 0, \quad P(X \leq E[X]) > 0.$$

Thm Let n, k, l , integers.

$$\text{Then } R(k, k) > n - \binom{n}{k} 2^{1 - \binom{k}{2}}$$

Pf: Take K_n , colour edges R/B each R w. prob $1/2$, B w. prob $1/2$; colours of edges indep.

Let A_i event i^{th} copy of K_k is monochr. $i=1, \dots, \binom{n}{k}$

$$\text{Let } X = \# \text{ monochr. } K_k = \sum_{i=1}^{\binom{n}{k}} I_{A_i}$$

$$\Rightarrow E[X] = \sum_{i=1}^{\binom{n}{k}} P(A_i) = \binom{n}{k} 2^{1 - \binom{k}{2}} =: \mu.$$

Since $P(X \leq \mu) > 0$, $\exists$ colouring of K_n s.t. $\leq \mu$ monochr. copies of K_k . Fix such a colouring.

Pick a vertex from each monochr. K_k .

Note, picked $\leq \mu$ many vert. Delete these vert.

Get R/B colouring of K_{ℓ} where $\ell \geq n - \mu$.
(no monochr. K_k)

$$\Rightarrow R(k, k) > \ell \geq n - \mu.$$

□

Cor $\forall \varepsilon > 0 \quad \exists k_0 \text{ s.t. } \forall k > k_0$

$$R(k, k) \geq (1 - \varepsilon) \frac{k^2}{e} \quad (\text{off by factor } \sqrt{2})$$

Pf: exercises.

Crossing Number

Recall Euler's formula G planar $\Rightarrow e(G) \leq 3|G| - 6$.

Thus $e(G) > 3|G| - 6 \Rightarrow$ any drawing of G on plane has crossing edges.

Defⁿ $cr(G)$ 'crossing # of G '.

smallest # crossings among all drawings of G where crossings of ≥ 2 edges @ one pt are not allowed. (Edges may be curved).

eg. K_4 planar $cr(K_4) = 0$



Note suppose G drawn s.t. #crossings = $cr(G)$.

Then • no edges crosses itself



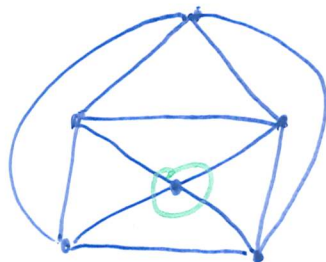
• edges uv, uw (common endpt) don't cross



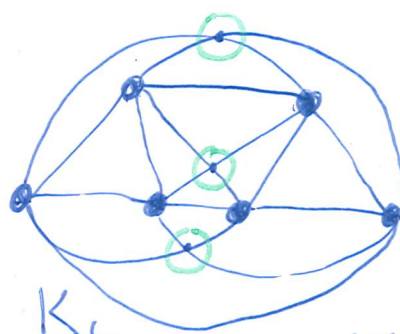
• no edges crosses twice



eg.



K_5 .



K_6 , $\binom{6}{2} = 15$ edges

simple
 given G , drawn with $cr(G)$ crossings

define H : $V(H) = V(G) \cup \{\text{all crossing pts}\}$

$E(H) \sim$ all pieces of original edges
 between vert. of H .

Note H is planar & simple (no loops
 no multi-edges)

• planar by defⁿ (no crossings).

• no loops



• no multi-edges



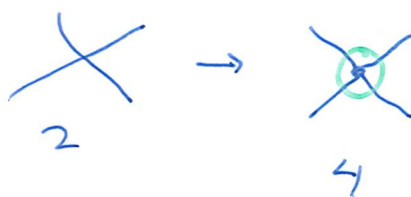
$\Rightarrow$ edge crosses itself twice.



$\Rightarrow$ edges with common endpoint in G cross.

Now, $v(H) = v(G) + cr(G)$

$e(H) = e(G) + 2 \cdot cr(G)$



$$\Rightarrow e(G) + 2 \cdot cr(G) \leq 3v(G) + 3cr(G) - 6$$

$$\Rightarrow cr(G) \geq e(G) - 3v(G) + 6$$

(*)

note $\Rightarrow cr(K_6) \geq 15 - 3 \cdot 6 + 6 = 3$.

Thm Let G simple graph, n vert. m edges, $m \geq 4n$

$$\Rightarrow cr(G) \geq \frac{1}{64} \cdot \frac{m^3}{n^2}$$

Pf

- Fix G , and minimal drawing of G .

Sample G_p : include each $v \in G$, indep. w. prob. p .

Let n_p, m_p, X_p # vert, edges, crossings in G_p .

$$\mathbb{E}(X_p - m_p + 3n_p) \geq 0$$

by (*)

Calc. $\mathbb{E}(n_p) = np$, $\mathbb{E}(m_p) = p^2 m$ (edge in G_p iff both endpts u, v, w, x)

$$\mathbb{E}(X_p) = p^4 cr(G)$$



(crossing pt. appears iff all 4 vert u, v, w, x)

$$\Rightarrow p^4 cr(G) - p^2 m + 3np \geq 0$$

$$\therefore cr(G) \geq \frac{p^2 m - 3np}{p^4} = \frac{m}{p^2} - \frac{3n}{p^4}$$

Take $p = \frac{4n}{m}$ (NB: $p \leq 1$ by assumption)

$$\Rightarrow cr(G) \geq \frac{1}{4^3} \left(\frac{4m}{(n/m)^2} - \frac{3n}{(n/m)^3} \right) = \frac{1}{64} \left(\frac{4m^3}{n^2} - \frac{3m^3}{n^2} \right) = \frac{1}{64} \frac{m^3}{n^2}$$

□

Asymptotic Notⁿ

Supp. f, g f^{ns}; $g(n) > 0 \quad \forall n \geq 1$.

$$\bullet f = O(g) \Rightarrow \exists C \text{ s.t. } |f(n)| \leq C \cdot g(n) \quad \forall n \geq 1.$$

$$\bullet f = o(g) \Rightarrow f(n)/g(n) \rightarrow 0 \text{ as } n \rightarrow \infty$$

$$\bullet f = \Theta(g) \Rightarrow f = O(g) \text{ and } g = O(f)$$

$$\text{i.e. } \exists c, c' \text{ s.t. } cg(n) \leq f(n) \leq c'g(n) \quad \forall n \geq 1.$$

$$\bullet f \sim g \Rightarrow \frac{f(n)}{g(n)} \rightarrow 1 \text{ as } n \rightarrow \infty.$$

$$\bullet f = \Omega(g) \Rightarrow g = O(f) \text{ i.e. } \exists c : f(n) \geq cg(n) \quad \forall n \geq 1.$$

examples $f = n^3 = O(n^2)$.

$$f = n^3 + O(n^2) \Rightarrow \exists C \text{ s.t. } f(n) \leq n^3 + Cn^2$$

$$f(n) \geq n^3$$

$$f = (1 + o(1))g \Leftrightarrow \forall \varepsilon > 0 \quad \exists n_0 \text{ s.t. } \forall n \geq n_0$$

$$\frac{|f(n) - g(n)|}{g(n)} \leq \varepsilon.$$